SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I K Sumanth Kumar Reddy (192425222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: K Sumanth

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
   Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing

During seasonal sales, an e-commerce company experiences sudden increases in website traffic. Cloud auto-scaling automatically adds or removes servers based on demand, while load balancing distributes incoming requests evenly across available servers.

Auto-Scaling

- Automatically increases server instances during high traffic.
- Removes extra instances when traffic decreases.
- Prevents server overload.

Load Balancing

- Distributes user requests across multiple servers.
- Improves response time.
- Prevents a single server from becoming overloaded.

Benefits

- Reduces downtime.
- Improves website performance.
- Provides a better user experience.
- Optimizes cloud resource usage and cost.

Role of Monitoring Tools

Monitoring tools continuously track CPU usage, memory, network traffic, and server health. If resource usage exceeds a threshold, they trigger auto-scaling and generate alerts for administrators.

Real-World Example

During the Amazon Great Indian Festival or Flipkart Big Billion Days, millions of customers visit the website simultaneously. Auto-scaling launches additional servers, while load balancing distributes customer requests efficiently, ensuring smooth shopping without delays.

2. Cloud Storage for Backup Data

Cloud storage provides a secure and reliable solution for storing large volumes of backup data.
Durability and Redundancy

- Data is automatically stored in multiple locations.
- Multiple copies protect against hardware failures.
- Replication ensures data is available even if one server fails.

Security

- Encryption protects data during storage and transmission.
- Access control allows only authorized users to access backup files.
- Multi-factor authentication (MFA) provides additional security.

Cost Advantages

- No need to purchase expensive storage hardware.
- Pay only for the storage used.
- Reduced maintenance and infrastructure costs.

Disaster Recovery Example
If a company's primary data center is damaged by a flood or fire, backup data stored in the cloud can be restored quickly, allowing business operations to continue with minimal downtime.

3. Content Delivery Network (CDN)

A **Content Delivery Network (CDN)** is a network of servers located around the world that delivers content from the server closest to the user.

Performance Improvement

- Reduces data travel distance.
- Loads websites and applications faster.
- Decreases network latency.

Edge Servers

- Store cached copies of frequently accessed content.
- Deliver data from nearby locations.
- Reduce the load on the main server.

Global Availability
Cloud providers deploy servers in multiple regions worldwide. Users are automatically connected to the nearest available server, ensuring fast and reliable service.

Example
Streaming platforms like **Netflix** or online games use CDNs to deliver videos and game data quickly to users worldwide, providing smooth streaming and low-latency gameplay.

4. Cloud Security and Compliance

Financial institutions require strong security and compliance when using cloud services.

Regulatory Compliance
Cloud providers support standards such as:

- ISO 27001
- PCI-DSS
- GDPR
- SOC 2

These standards help organizations meet legal and security requirements.

Logging and Monitoring

- Record all user activities.
- Detect suspicious behavior.
- Help during security audits and investigations.

Encryption

- Encrypts data at rest and in transit.
- Protects sensitive financial information from unauthorized access.

Data Sovereignty
Organizations can choose cloud regions where data is stored to comply with local government regulations and privacy laws.

Example
During an online banking transaction, customer information and payment details are encrypted, all activities are logged, and the data is stored in an approved regional data center to meet compliance requirements.

5. Containers and Kubernetes

Containers package an application and its dependencies so it can run consistently across different environments.

Containers vs Virtual Machines

Containers	**Virtual Machines**
Lightweight	Heavier
Faster startup	Slower startup
Share host OS	Separate operating system
Lower resource usage	Higher resource usage

Kubernetes Orchestration

Kubernetes simplifies:

- Automatic deployment
- Load balancing
- Auto-scaling
- Self-healing of failed containers
- Centralized management

Portability

Containers run consistently across different cloud platforms such as AWS, Microsoft Azure, and Google Cloud, reducing compatibility issues.

Example

An **online shopping application** can use microservices such as:

- User Service
- Product Service
- Order Service
- Payment Service
- Notification Service